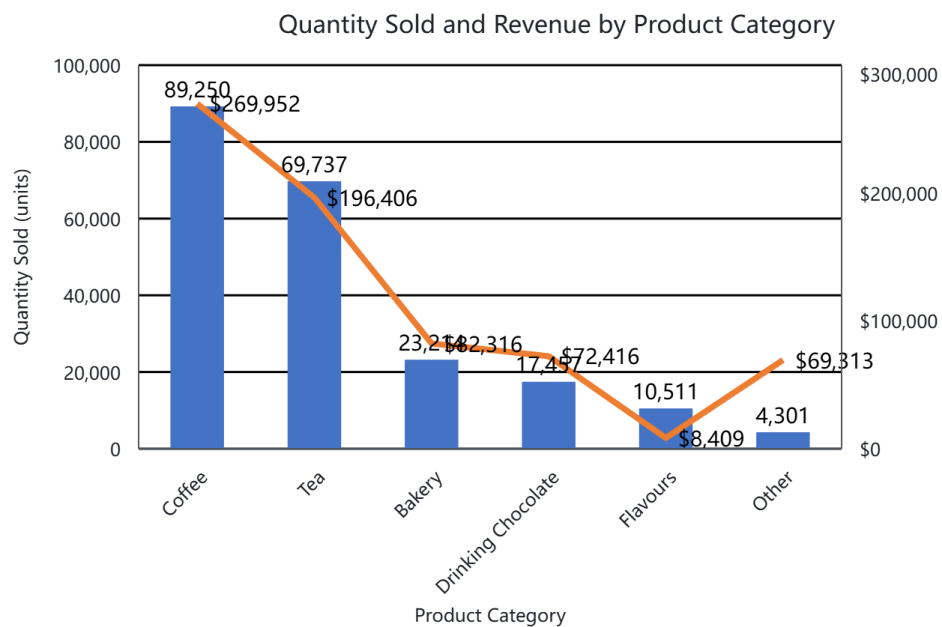
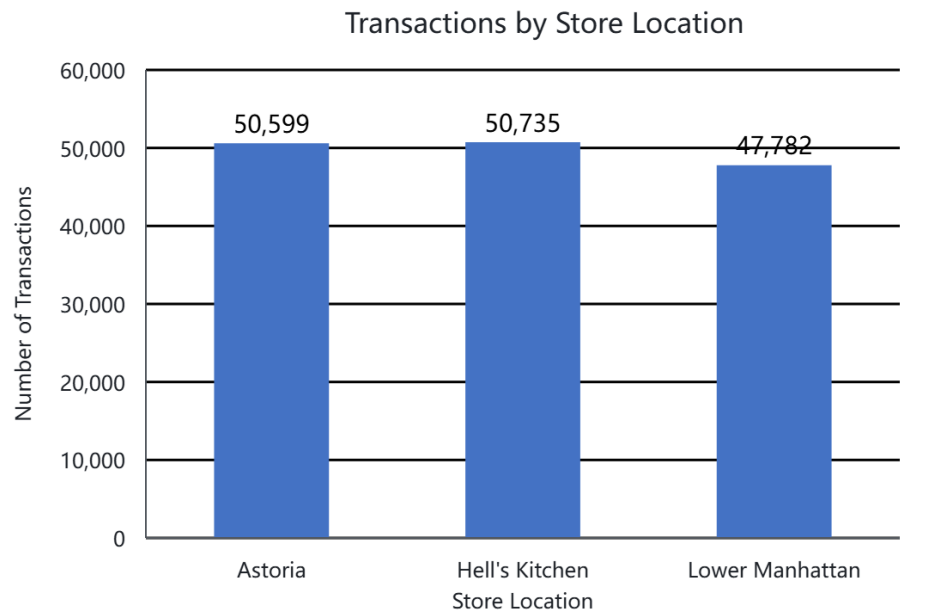


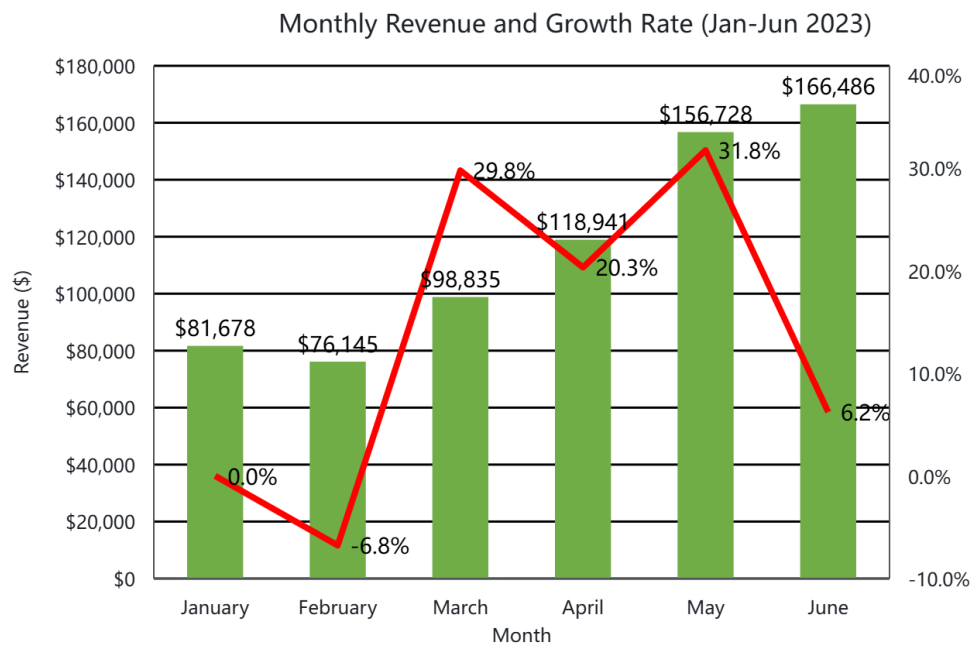
## Analysis Report

After opening the data, I found it covered three locations: Hell's Kitchen, Astoria, and Lower Manhattan, with a total of 149,116 transactions from January 1 to June 30, 2023. I first analyzed transaction volumes by store: Astoria had 50,599 transactions, Hell's Kitchen had 50,735, and Lower Manhattan had 47,782. The numbers were nearly identical across all three stores, indicating no significant difference in sales performance. Location therefore does not appear to be a major factor affecting sales. Next, I broke down sales by product category, Coffee accounted for 89,250 units (41.6%), tea for 69,737 units (32.5%), pastries for 23,214 units (10.8%), hot chocolate for 17,457 units, syrups for 10,511 units, and all other items combined were under 2,000 units. This clearly shows that coffee is the dominant product. Then, I reviewed monthly revenue: \$81,678 in January, slightly lower at \$76,145 in February, rising to \$98,835 in March, \$118,941 in April, \$156,728 in May, and reaching \$166,486 in June. Revenue more than doubled from February to June, with four consecutive months of growth, signaling a period of rapid business expansion. Additionally, I noticed that beverages sold 176,444 units compared to only 23,214 units for pastries. A sevenfold difference. Based on these findings, I offer three recommendations. First, since coffee accounts for 40% of revenue, we should focus on introducing new coffee products. Second, given the strong growth momentum, now is an ideal time to launch new items or consider expansion. Third, the large gap between beverage and pastry sales suggests creating combo packages that pair drinks with pastries, leveraging beverage traffic to boost pastry sales.

After completing this assignment, I have two questions. First, I noticed that under each `product_type` there are many more detailed `product_details`, totaling 80 types. Manually

counting the sales volume for each product\_detail would be extremely time-consuming. I'd like to know if there's an efficient way to do this quickly. Second, the data includes both store\_id and store\_location, which correspond one-to-one. I'm wondering why both fields are kept.





I found that visualizing the data made the gaps between stores, categories, and months much clearer than text alone, so I included these charts to support my observations.